Performance Test

Test Run ID: 3fa83ab8-8f8b-458c-9b94-2e23624d6dd5

This chapter provides a summary of the test run metrics. The tables contain the aggregated values of the metrics for the entire test run.

Date	08 June 2026	**Max VUs Tested**	80
Executor	constant-vus	**Tool**	k6

1. Executive Summary

The target endpoint for this performance test is the "https://quickpizza.grafana.com" URL. The testing type is a constant load scenario, where 80 Virtual Users (VUs) are maintained throughout the 45-second test duration. The load profile consists of a constant arrival rate of 80 VUs, with no ramping or stages, indicating a steady-state load test.

Metric	Value
Minimum TPS	40.00
Average TPS	60.85
Maximum TPS	159.50

2. Performance Summary

Trends

metric	avg	max	med	min	p90	p95	p99
http_req_duration	276.89ms	578.30ms	261.39ms	245.72ms	322.66ms	328.54ms	-
http_req_waiting	276.57ms	578.10ms	261.25ms	245.62ms	321.98ms	327.79ms	-
http_req_connecting	7.56ms	279.91ms	0.00ms	0.00ms	0.00ms	0.00ms	-
iteration_duration	1294.92ms	1914.50ms	1263.31ms	1247.26ms	1327.88ms	1354.92ms	-

Counters

metric	count	rate
data_received	9.42 MB	208.48 kB/s
data_sent	282.43 kB	6.10 kB/s
http_reqs	2,816	60.85/s
iterations	2,816	60.85/s

Rates

metric	rate
checks_succeeded	100.00% (2816/2816)
checks_failed	0.00% (0/2816)
http_req_failed	0.00% (0/2816)

Gauges

metric	value
vus	16 (min: 16, max: 80)
vus_max	80 (min: 80, max: 80)

Checks

Check Name	Passed	Failed	Success Rate	Status
status is 200	2816	0	100.00%	Pass

3. Advanced Analysis

3.1 Error Analysis

• **Network/Connection errors**: None detected.

• **Application errors**: None detected.

• **Infrastructure errors**: None detected.

• **Warnings**: 1 warning detected in the logs, indicating unknown fields in the options exported in the script.

3.2 Observation

The performance test results indicate a stable system with no significant errors or failures. However, upon closer inspection of the logs, a warning message is observed indicating unknown fields in the options exported in the script. This warning is likely due to a minor issue with the test configuration.

The test results also show a consistent average duration of 276.59ms for HTTP requests, with a maximum duration of 578.30ms. This suggests that the system is able to handle requests within a reasonable time frame.

The iteration duration shows an average of 1294.92ms, with a maximum of 1914.50ms. This indicates that the system is able to complete iterations within a reasonable time frame.

The VU (Virtual User) count remains stable at 16 throughout the test, indicating that the system is able to handle the load without any issues.

3.3 Final Recommendations

1. **Advanced Connection Management**: Implement a connection pooling mechanism to reduce the overhead of establishing new connections for each request.

2. **Session Persistence sticky affinity/Redis cache**: Implement a session persistence mechanism to reduce the overhead of re-establishing sessions for each request.

3. **Gateway resilience Istio outlier circuit breaker**: Implement an outlier detection mechanism to detect and prevent cascading failures in the system.

4. **Observability tracing and standardized log levels**: Implement a tracing mechanism to provide detailed insights into the system's behavior and standardize log levels to improve log analysis.

5. **Database read/write splitting and PgBouncer connection pooling**: Implement a read/write splitting mechanism to reduce the load on the database and implement a connection pooling mechanism to reduce the overhead of establishing new connections for each request.

3.4 Conclusion

The performance test results indicate a stable system with no significant errors or failures. However, the warning message in the logs indicates a minor issue with the test configuration. The system is able to handle requests within a reasonable time frame, and the iteration duration is also within acceptable limits.

Based on the test results and analysis, the system has passed its performance objectives. However, to further improve the system's performance and reliability, the recommended architectural and database optimization changes should be implemented.